

2501.06662: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2501.06662_E00051	21	■ 0.536	$\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t. \end{aligned}$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$
2501.06662_E00059	23	■ 0.550	$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n} &= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \perp) \log \pi(\perp a_1 \cdots a_{N-1} \perp) \end{aligned}$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n} = \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \perp) \log \pi(\perp a_1 \cdots a_{N-1} \perp)$	$\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n} = \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1} \perp) \log \pi(\perp a_1 \cdots a_{N-1} \perp)$
2501.06662_E00058	23	■ 0.594	$\begin{aligned} \mathcal{H}(\mathcal{M}, p, \zeta) &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right) \\ &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z y)^t p(z) \right). \end{aligned}$	$\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z y)^t p(z) \right).$	$\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left(\sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z y)^t p(z) \right).$
2501.06662_E00023	15	■ 0.636	$\begin{aligned} \#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} &= \sum_{i=0}^{N-2} \#A^i \\ &= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1. \end{aligned}$	$\#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i\right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1.$	$\#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i\right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1.$
2501.06662_E00052 (13)	21	■ 0.693	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$	$\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$

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2501.06662_E00033	17	■ 0.720	$\zeta_{\mathbf{t}}^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^{\mathbf{t}} & \text{if } y \in \mathbf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathbf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$	$\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathbf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$
2501.06662_E00037 (10)	18	■ 0.780	$\begin{aligned} \operatorname{Mag}(\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{\mathbf{t}}^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_{\mathbf{t}}^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathbf{L}_x^{(1)}} \pi(y \mid x)^{\mathbf{t}} \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^{\mathbf{t}} \right) + \#(T(\perp)). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(t\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathbf{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(t\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathbf{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$
2501.06662_E00049	20	■ 0.797	$\begin{aligned} \partial_k^i(y_0, \dots, y_k) &= \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases} \end{aligned}$	$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$	$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$
2501.06662_E00001	3	■ 0.822	$\operatorname{Mag}(\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$
2501.06662_E00029	16	■ 0.829	$\left(\zeta_{\mathbf{t}} - \delta \right)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^{\mathbf{t}}.$	$(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$	$(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$
2501.06662_E00036 (11)	18	■ 0.835	$\operatorname{Mag}(\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$	$\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$

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2501.06662_E00026 (8)	15	■ 0.842	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$	$\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i y_{i-1})^t.$
2501.06662_E00021	14	■ 0.849	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1). \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1) \end{aligned}$	$\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1). \end{aligned}$
2501.06662_E00022	15	■ 0.871	$\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$	$\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2 \end{aligned}$	$\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$
2501.06662_E00045	19	■ 0.922	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)},$	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)},$
2501.06662_E00039 (12)	18	■ 0.940	$f'(\text{\textbackslash prime})(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$	$f'(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$

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2501.06662_E00046	20	■ 0.942	$\begin{aligned} \operatorname{Mag}(\mathcal{M}_x) &= \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ &= \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in \mathcal{L}_y^{(1)}} \pi(z y)^t \\ &= (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)). \end{aligned}$	$\operatorname{Mag}(\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in \mathcal{L}_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$	$\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in \mathcal{L}_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$
2501.06662_E00040	19	■ 0.945	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M})} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$	$f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$
2501.06662_E00044	19	■ 0.971	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$	$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$
2501.06662_E00032	17	■ 0.973	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$	$\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$
2501.06662_E00056	22	■ 0.986	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$	$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$
2501.06662_E00015 (7)	12	■ 0.988	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$	$\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$
2501.06662_E00002	4	■ 0.991	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$	$\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$
2501.06662_E00005	6	■ 0.991	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$	$\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and } a \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and } a < N-1 \}.$
2501.06662_E00006	8	■ 0.996	$\begin{aligned} T(x) &= \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ &\quad y \text{ is a finished text such that } y \leq N \}. \end{aligned}$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that } y \leq N \}.$	$T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that } y \leq N \}.$
2501.06662_E00034	17	■ 0.998	$\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$	$\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$	$\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$

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2501.06662_E00003	4	■ 0.998	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \text{Mag}(t\mathcal{M}) \leq \underbrace{\#(\text{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \text{Mag}(t\mathcal{M}) \leq \underbrace{\#(\text{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$	$\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \text{Mag}(t\mathcal{M}) \leq \underbrace{\#(\text{ob}(\mathcal{M}))}_{\text{maximally random LM}}.$
2501.06662_E00054 (14)	22	■ 0.998	$\text{Mag}(t\mathcal{M}) = \text{rank } H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \text{rank } H_{1,\ell}(\mathcal{M}).$	$\text{Mag}(t\mathcal{M}) = \text{rank } H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \text{rank } H_{1,\ell}(\mathcal{M}).$	$\text{Mag}(t\mathcal{M}) = \text{rank } H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \text{rank } H_{1,\ell}(\mathcal{M}).$
2501.06662_E00019	14	■ 0.999	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}.$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}.$	$\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}.$
2501.06662_E00042	19	■ 0.999	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$	$H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1},$
2501.06662_E00004	5	■ 1.000	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M})),$	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M})),$	$\text{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \text{rank}(H_{k,\ell}(\mathcal{M})),$
2501.06662_E00012	11	■ 1.000	$\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$	$\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}. \end{aligned}$
2501.06662_E00031	17	■ 1.000	$\begin{aligned} & \prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ & = \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t \\ & = \vdots \\ & = \pi(y_k y_0)^t \end{aligned}$	$\prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$ $= \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$ $= \vdots$ $= \pi(y_k y_0)^t$	$\prod_{i=1}^k \pi(y_i y_{i-1})^t = \pi(y_1 y_0)^t \pi(y_2 y_1)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$ $= \pi(y_2 y_0)^t \pi(y_3 y_2)^t \cdots \pi(y_k y_{k-1})^t$ $= \vdots$ $= \pi(y_k y_0)^t.$
2501.06662_E00007	8	■ 1.000	$p(a_3 \perp a) p(a_4 \perp aa_3) p(a_5 \perp aa_3 a_4) p(\dagger \perp aa_3 a_4 a_5).$	$p(a_3 \perp a) p(a_4 \perp aa_3) p(a_5 \perp aa_3 a_4) p(\dagger \perp aa_3 a_4 a_5).$	$p(a_3 \perp a) p(a_4 \perp aa_3) p(a_5 \perp aa_3 a_4) p(\dagger \perp aa_3 a_4 a_5).$

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2501.06662_ EQ0008 (1)	9	■ 1.000	$\pi(y \mid x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$	$\pi(y \mid x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$	$\pi(y x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$
2501.06662_ EQ0009 (2)	9	■ 1.000	$\sum_{y \in T(x)} \pi(y \mid x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$	$\sum_{y \in T(x)} \pi(y \mid x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$	$\sum_{y \in T(x)} \pi(y x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$
2501.06662_ EQ0010 (4)	9	■ 1.000	$\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a' a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$	$\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a' a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$	$\begin{aligned} \sum_{y \in T(x)} \pi(y x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a' a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$
2501.06662_EQ0011	10	■ 1.000	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) &\otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$

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2501.06662_E00013 (6)	11	■ 1.000	$\begin{aligned} \pi(y \mid x) \pi(z \mid y) &= \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j}) \\ &= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i}) \\ &= \pi(z \mid x). \end{aligned}$	$\pi(y \mid x) \pi(z \mid y) = \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i})$ $= \pi(z \mid x)$	$\pi(y x)\pi(z y) = \prod_{i=1}^k p(a_{t+i} y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} z_{<t+i})$ $= \pi(z x).$
2501.06662_E00014	12	■ 1.000	$d(x, y) := -\ln \pi(y \mid x) .$	$d(x, y) := -\ln \pi(y \mid x).$	$d(x, y) := -\ln \pi(y x).$
2501.06662_E00016	13	■ 1.000	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$	$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$
2501.06662_E00017	13	■ 1.000	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$	$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$
2501.06662_E00018	13	■ 1.000	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P .$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$	$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$
2501.06662_E00020	14	■ 1.000	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$	$\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$
2501.06662_E00024	15	■ 1.000	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$	$\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y x),$
2501.06662_E00025	15	■ 1.000	$\left(\zeta_{\mathcal{M}} \right)_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t .$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t .$	$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y x)^t .$
2501.06662_E00027 (9)	16	■ 1.000	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$	$\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k,$
2501.06662_E00028	16	■ 1.000	$\left(\zeta_t - \delta \right) (x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$	$(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$
2501.06662_E00030	16	■ 1.000	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$	$\zeta_t^{-1}(x, y) = \pi(y x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$
2501.06662_E00035	17	■ 1.000	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right) .$	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right) .$	$H_t(p) = \frac{1}{t-1} \left(1 - \sum_{i=1}^n p_i^t \right) .$

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2501.06662_E00038	18	■ 1.000	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$	$\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$
2501.06662_E00041	19	■ 1.000	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t \ln p_x(a)}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$	$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$
2501.06662_E00043	19	■ 1.000	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$	$\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$
2501.06662_E00047	20	■ 1.000	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$	$G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$
2501.06662_E00048	20	■ 1.000	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$	$\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$
2501.06662_E00050	21	■ 1.000	$\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$	$\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$	$\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$
2501.06662_E00053	21	■ 1.000	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$	$\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \#G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$
2501.06662_E00055	22	■ 1.000	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$	$PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i y_{<i}) \right\}$
2501.06662_E00057	23	■ 1.000	$\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$	$\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$	$\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$
2501.06662_E00060	23	■ 1.000	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$	$-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$